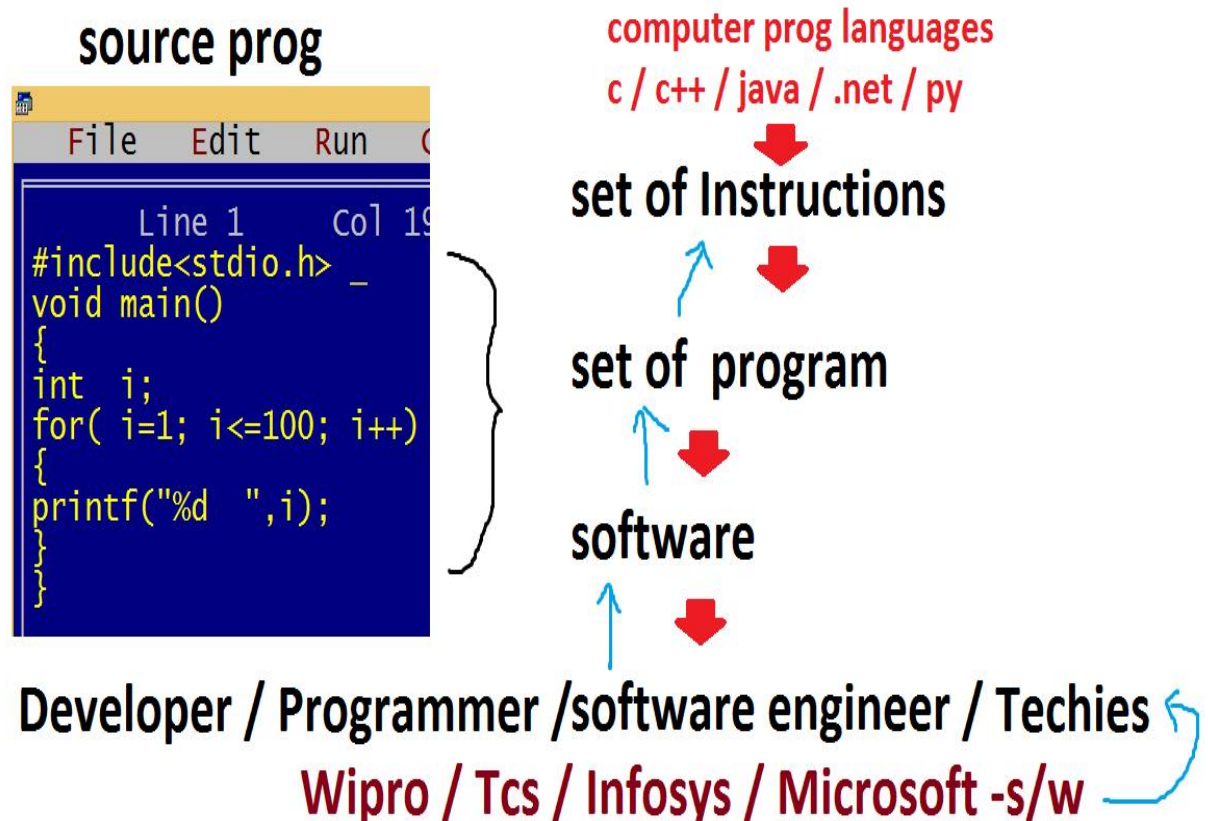


INTRODUCTION TO C

C is a

1. It is a high level / middle level programming language.



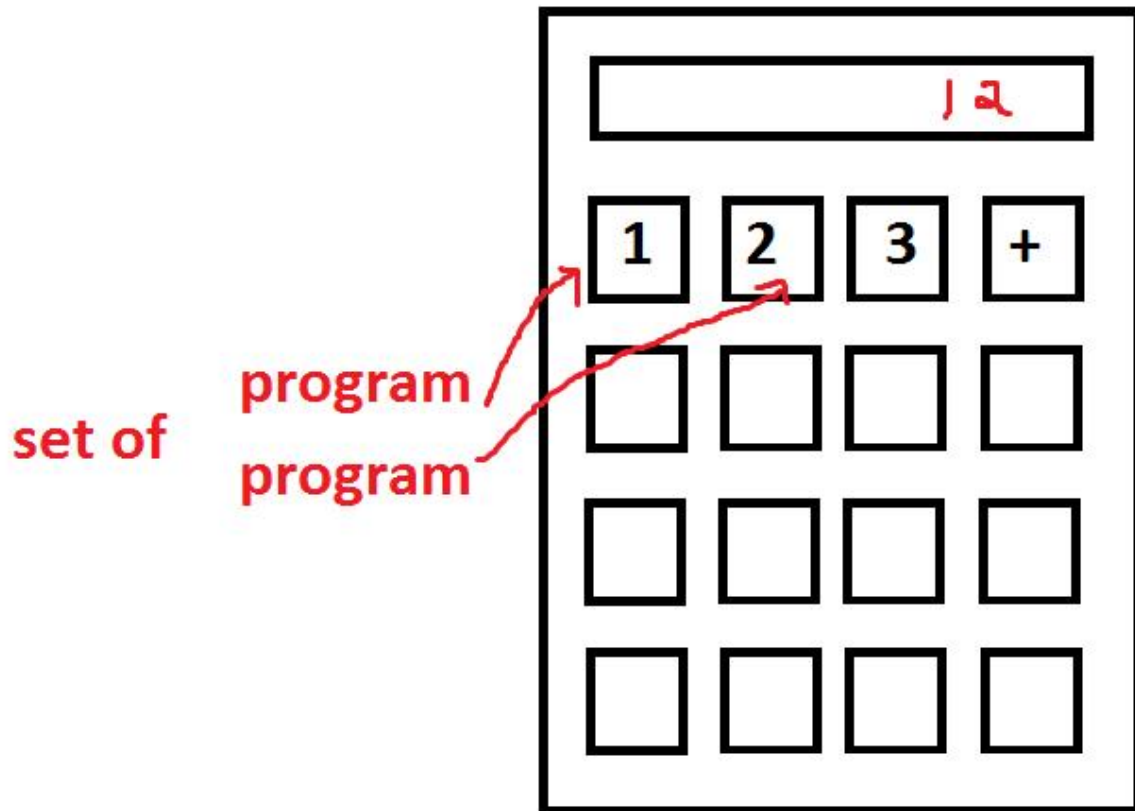
What is a program?

Set of instructions is called program.

What is a software?

Set of programs is called software. Or As per the IT industry software is a digitalized and automated process.

Calculator Software



2Hours - Timer - Automatically The A/C Power Off

Basically the software divided into 2 types.

1. System software

Eg: operating system, device drivers, translators

2. Application software

Eg: phonepe, whatsapp, instagram, irctc,.....

What is a language?

Generally the languages like telugu / English / hindi etc are called human languages, which are used to **communicate** with humans. The languages like C / C++ / Java / .Net / Python etc are called **computer programming languages**, which are used to develop the programs [**software**] to communicate with the machines to make our work efficient and accurate.

Basically the languages are divided into 3 types.

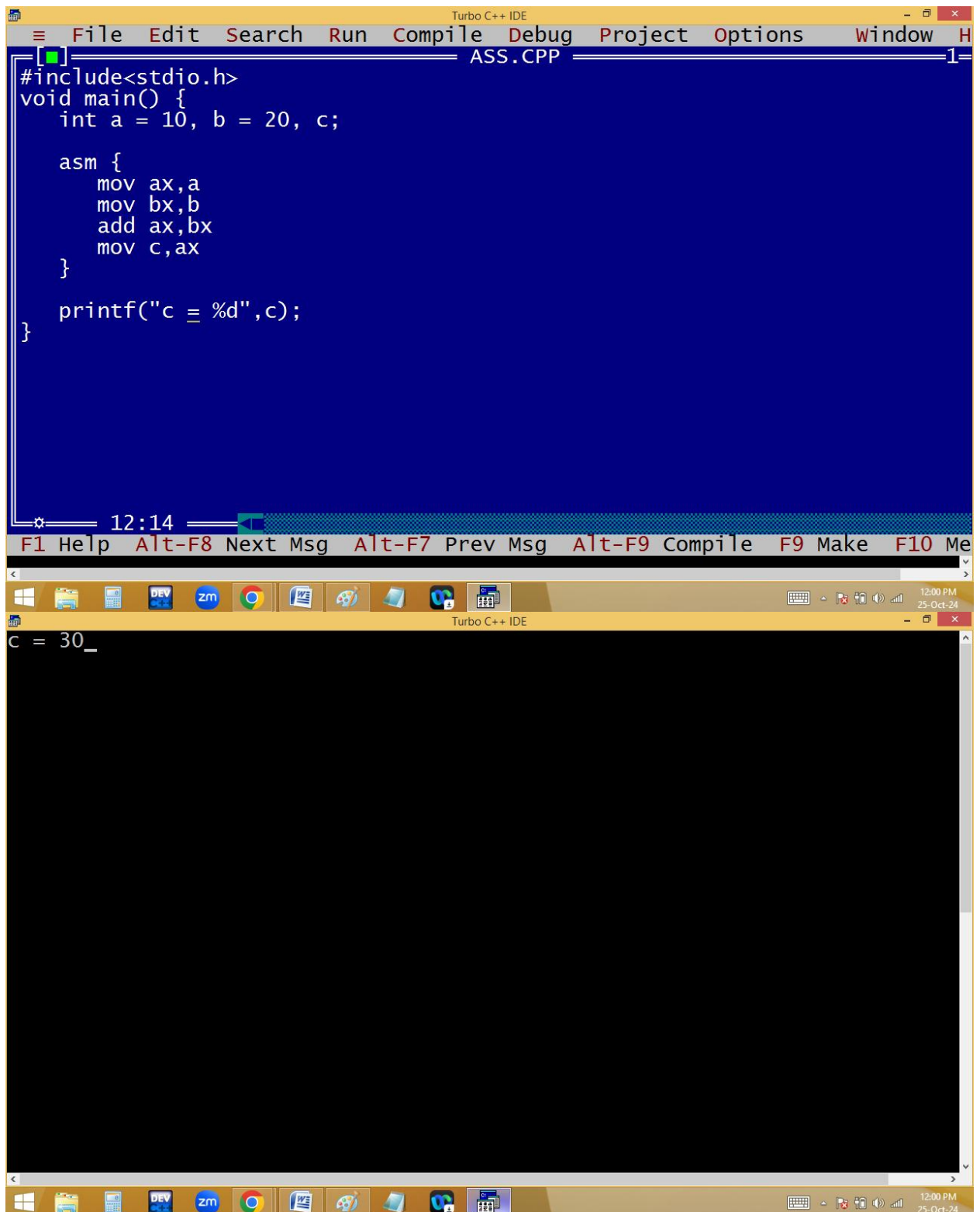
1. **Machine language**: It is created with **binary code**[0,1]

Eg: 10001111

2. **Low level / assembly language**: Created with English like shortcuts called **MNEMONICS**.

Eg: add, sub,...

Example for low / assembly language:



The image shows a screenshot of the Turbo C++ IDE. The top window, titled 'ASS.CPP', contains the following C code:

```
#include<stdio.h>
void main() {
    int a = 10, b = 20, c;

    asm {
        mov ax,a
        mov bx,b
        add ax,bx
        mov c,ax
    }

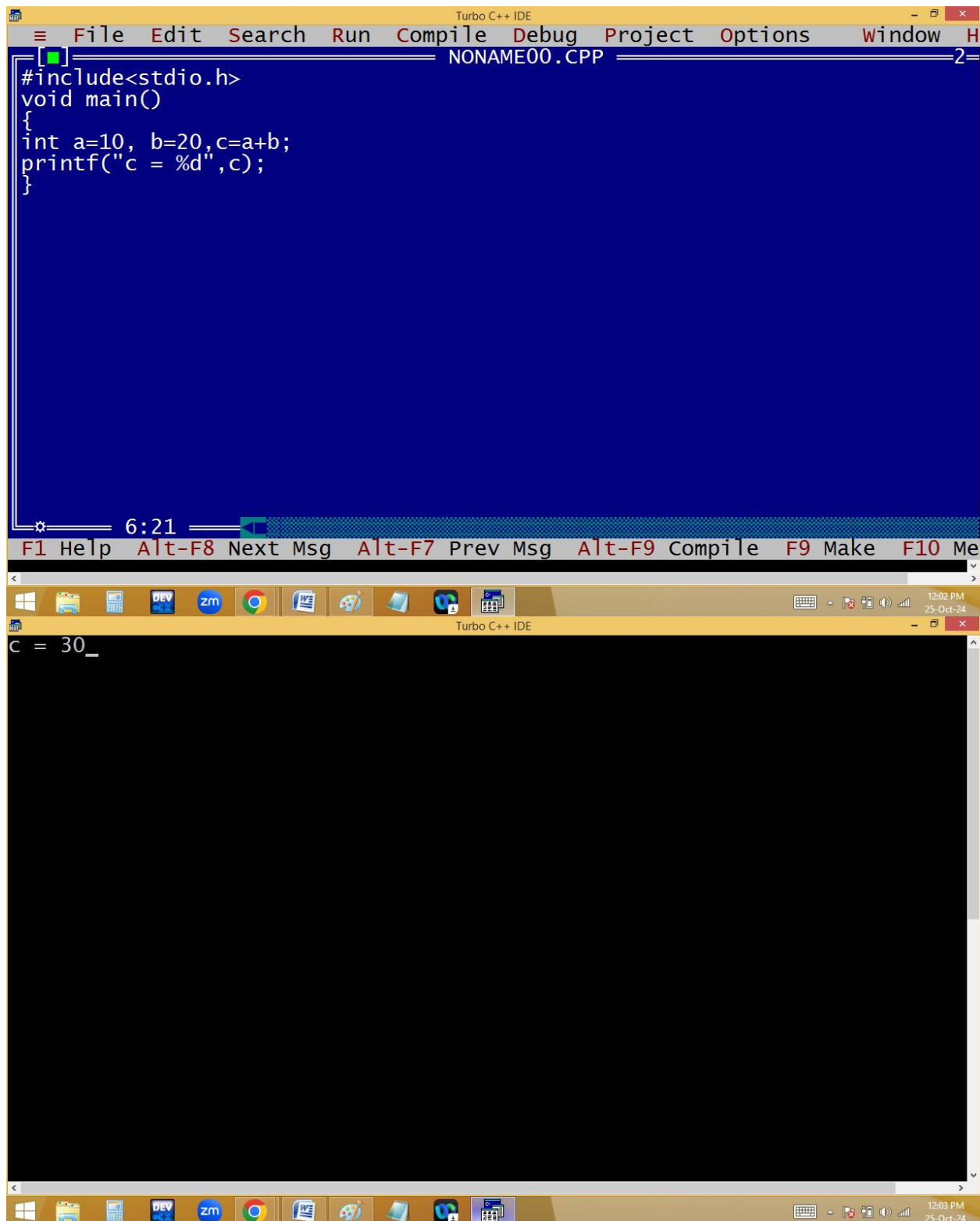
    printf("c = %d",c);
}
```

The bottom window shows the output of the program: 'c = 30_'. The IDE interface includes a menu bar (File, Edit, Search, Run, Compile, Debug, Project, Options, Window, Help), a toolbar, and a status bar at the bottom with the time '12:00 PM' and date '25-Oct-24'.

3. **High level language:** Created with **simple English**. Hence they are easy to understand.

Eg: addition, subtraction,....

Example for high level language:



The image shows a screenshot of the Turbo C++ IDE. The top window, titled 'NONAME00.CPP', contains the following C code:

```
#include<stdio.h>
void main()
{
int a=10, b=20,c=a+b;
printf("c = %d",c);
}
```

The bottom window shows the output of the program: 'c = 30_'. The IDE's status bar at the bottom indicates the time as 12:02 PM on 25-Oct-24.

C comes with both low level and high level features. Hence c is also called it is a middle level programming language.

The **C** low level features used to develop system software and high level features used to develop application software. Hence **c** is also called it is a **multi-purpose** programming language.

2. **C** is a compiler based programming language.

What is a translator?

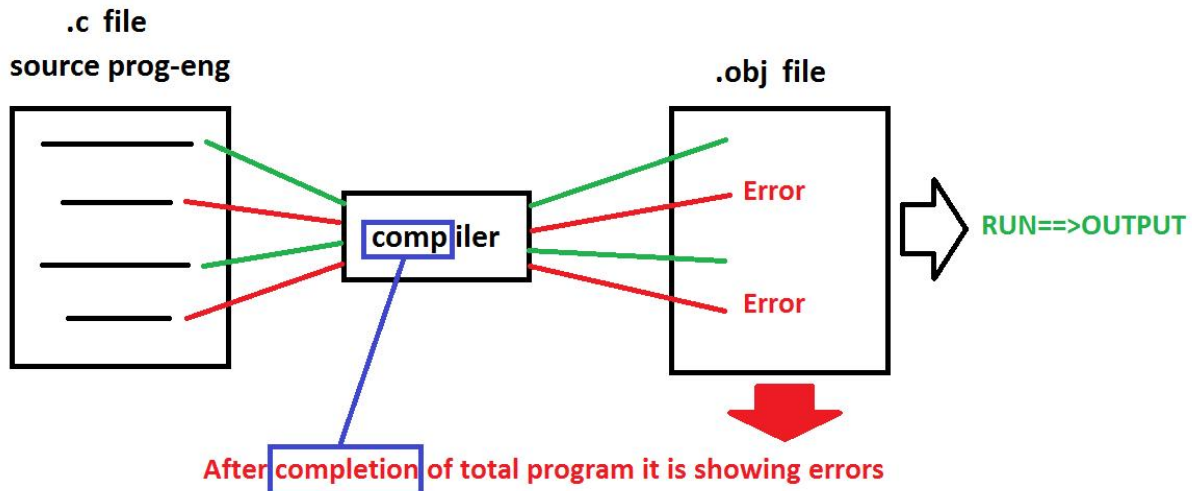


Always the programmer / user given instructions are in English, which is called source code / source program. But the computer understandable code is machine language / binary code. To **convert this source code into binary code and to check the programming mistakes [Errors]** we are using **translators** like

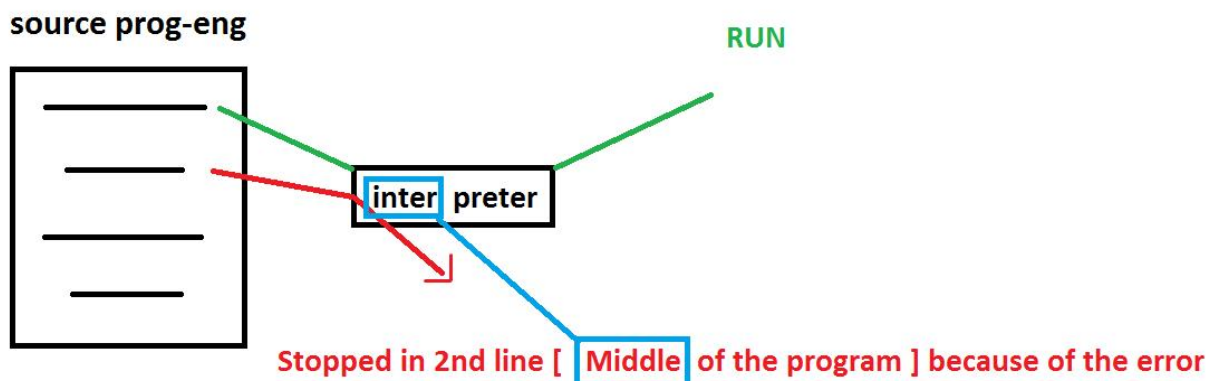
1. Compiler
2. Interpreter
3. Assembler

Compiler and interpreter used to convert **high level** programs to **machine language**.

Compiler converts the total program / source code into binary code at once by leaving error lines.



Interpreter converts line by line.



Assembler working style is similar to compiler.

Assembler used to convert **low level** programs to **machine language**.

In C we are using compiler as a translator. Hence it is called compiler based programming language.

In java we are using compiler and interpreter. Hence it is called compiler based interpreted language.

3. C is a procedure oriented programming language [POP's].

What is programming paradigm?

Every programming language comes with a particular structure with rules and regulations, which is technically called **programming paradigm**.

Before C language, the languages are using **Monolithic** programming structure, which comes with single program. Due to this it is very difficult to identify the errors, take more memory and execution speed is less. In monolithic programming we can't use part of the program when it is required and every time we have to run the entire program.

The image shows a screenshot of the Turbo C++ (TC) IDE. The top window is the editor, displaying a C program. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the bottom of the editor shows "Line 1 Col 44 Insert Indent Tab Fill Unindent * E:11AM.C". The code in the editor is as follows:

```
/* source program - monolithic programming */
#include<stdio.h>
void main()
{
printf("-----\n");
printf("Good morning\n");
printf("-----\n");
printf("welcome to c\n");
printf("-----\n");
printf("Thank you\n");
printf("-----");
}
```

Below the editor is a "Watch" window, which is currently empty. At the bottom of the IDE is a toolbar with various function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, F10-Menu, and NUM SC. The bottom-most window is the output window, which shows the execution results of the program:

```
-----
Good morning
-----
welcome to c
-----
Thank you
-----
```

The Windows taskbar is visible at the very bottom of the screen, showing the Start button, several application icons (including DEV, zm, Chrome, Word, and the TC icon), and the system clock indicating 11:44 AM on 28-Oct-24.

What is POP's?

In Procedure oriented programming structure, the big programs are divided into several small sub programs / sub routines / procedures / functions / modules / structures. Hence c program is collection of procedures, it is called POP's.

Advantages:

1. **Modularity:** Dividing big program into small pieces according to the program requirement.
2. **Reusability:** Write once, use many times.
3. **Simplicity:** Easy to understand the program
4. **Efficiency:** Performance is high.

The image shows a screenshot of the Turbo C++ (TC) IDE. The top window is the 'Edit' window, which contains a C program. The program defines a function 'line()' that prints a line of dashes, and a 'main()' function that calls 'line()' four times to print 'Good morning', 'Welcome to C', and 'Thank you'. The bottom window is the 'Watch' window, which displays the output of the program: three lines of dashes, followed by 'Good morning', 'Welcome to C', and 'Thank you', each followed by a line of dashes. The Windows taskbar at the bottom shows the time as 12:43 PM on 28-Oct-24.

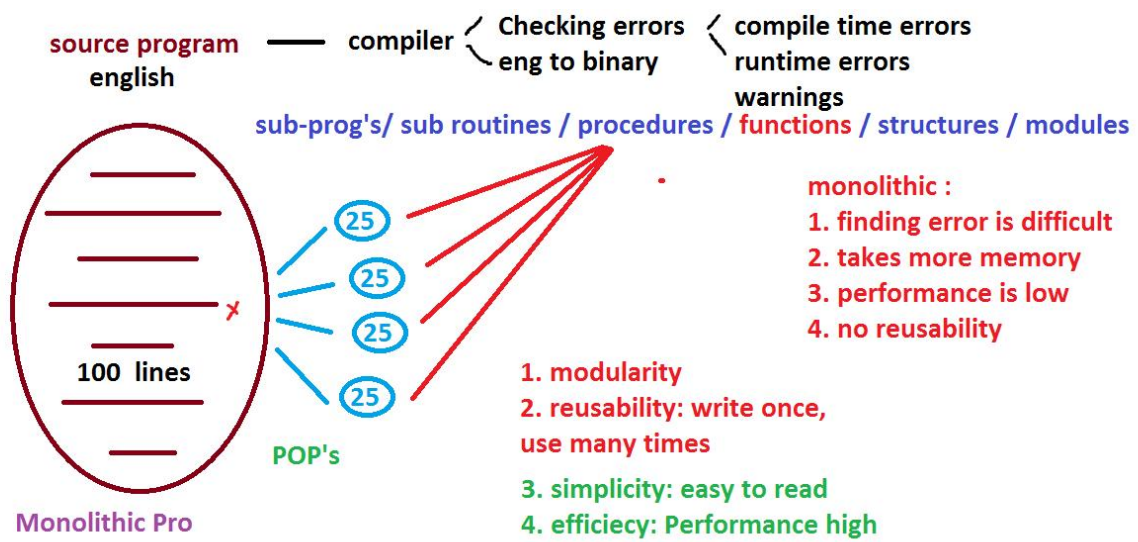
```
Line 1 Col 8 Insert Indent Tab Fill Unindent * E:11AM.C
/* source program - Procedure oriented programming */
#include<stdio.h>
void line( )
{
printf("-----\n");
}
void main()
{
line();
printf("Good morning\n");
line();
printf("welcome to C\n");
line();
printf("Thank you\n");
line();
}
```

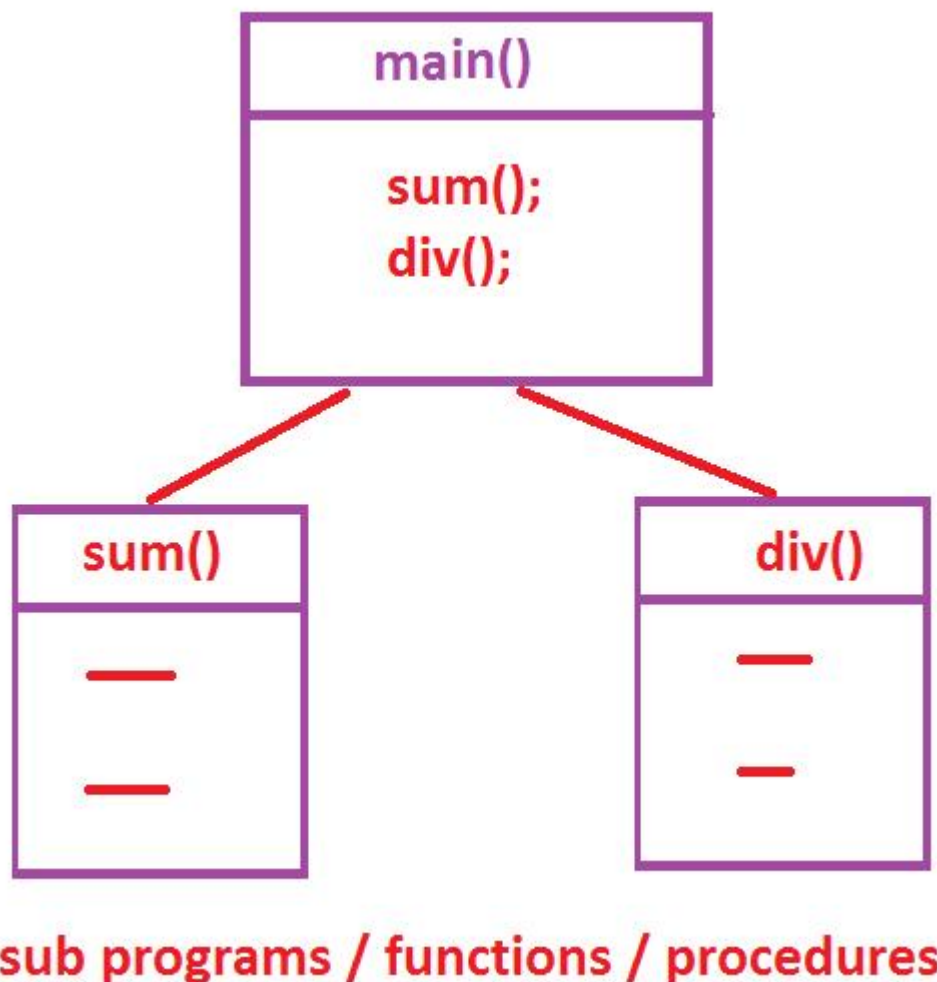
Good morning

Welcome to C

Thank you

2





POP Drawback:

Data is not secured because of data is public.

OOP's: Object Oriented Programming structure

Features:

- 1. Class**
- 2. Object**
- 3. Data hiding**

- 4. Encapsulation
- 5. Inheritance
- 6. Polymorphism
- 7. Abstraction etc

C supports Only the POP's

C++ and **Python** having **POP's** and **OOP's**. hence they are called **multi paradigm** programming language.

Java supports only OOP's.

HOTEL PARADISE BEGUMPET-HYD				
ITEM		QTY	PRICE	AMOUNT
Object1 —	A	1	280	280
Object2 —	B	1	380	380
Object3 —	C	1	30	30
TOTAL: AAAAA				

4. C is a General purpose / multi-purpose programming language.

Using C language we can develop different type of applications / software like

1. Operating systems

Eg: windows, unix, linux, android, mac,...

2. Editors

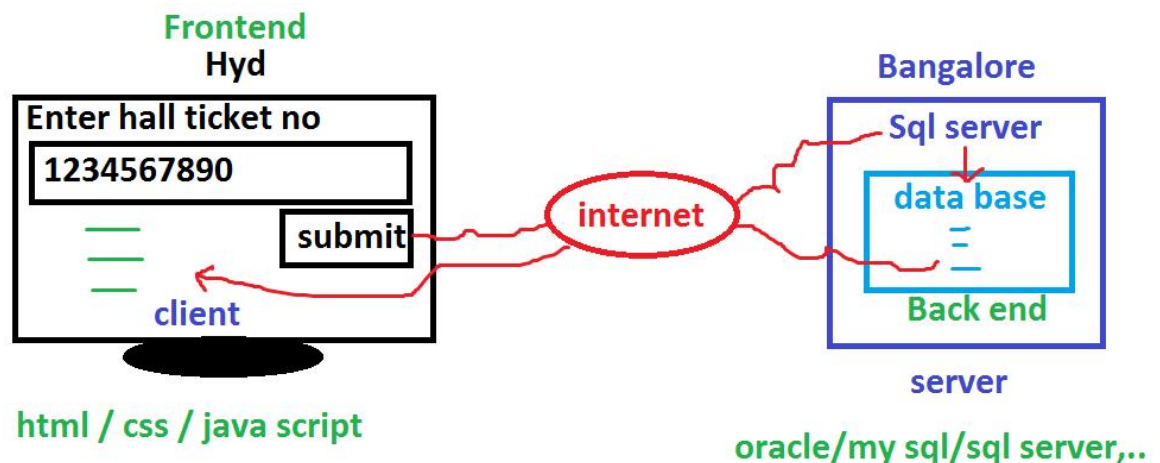
Eg: Ms-Word, Notepad, Wordpad, Edit+,...

3. Translators

Eg: compiler, interpreter, assembler

4. Data base

Eg: Oracle, My SQL, SQL Server, Mongo DB, DBII,....



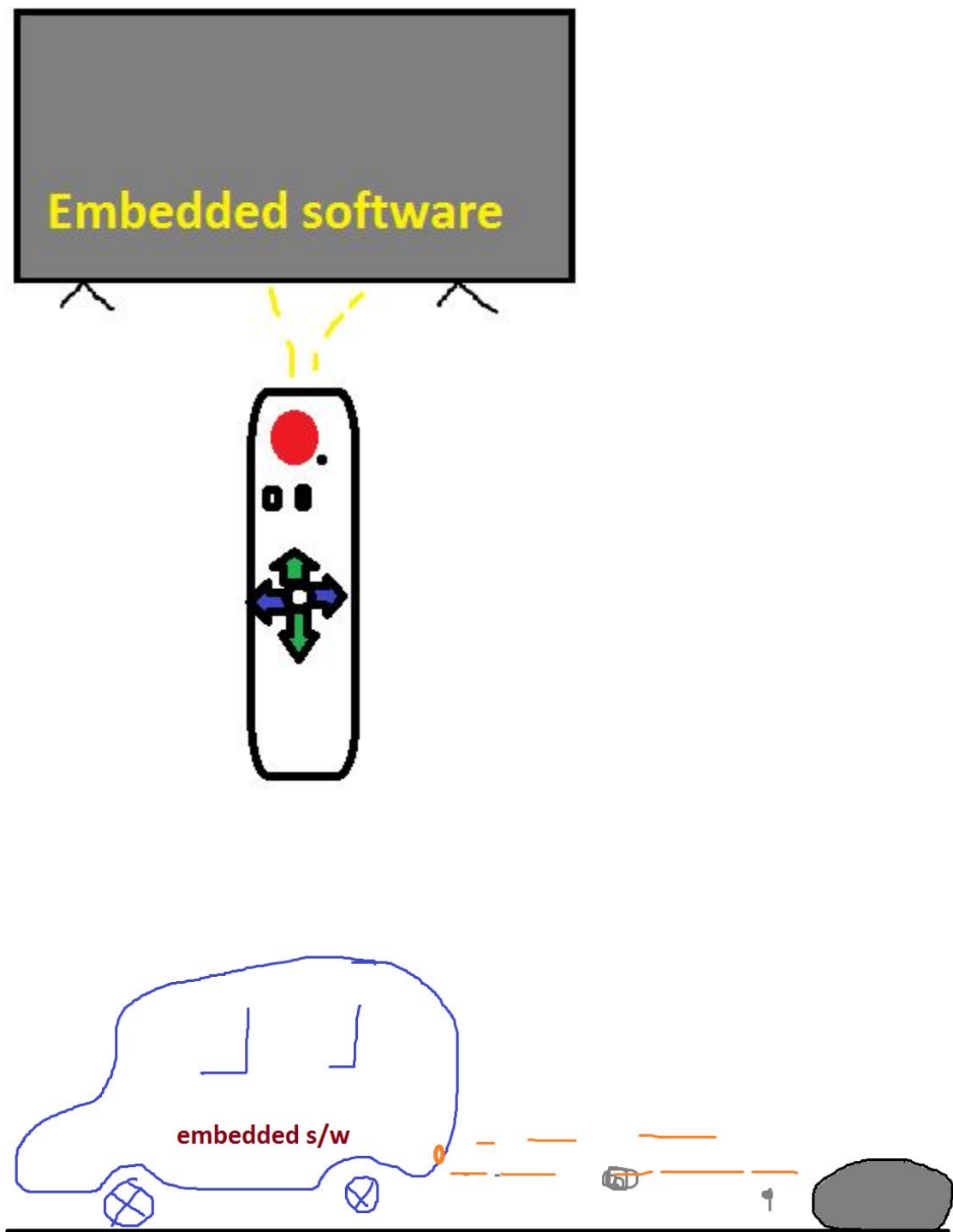
5. Commercial applications

Eg: hotel / college / super market / atm ,....

6. Device drivers

Eg: audio / video / printer / usb drivers etc.

7. Embedded applications



8. Anti virus

Eg: avast, macfee, nod,...

9. Media players

Eg: vlc, mx-player, windows media player,....

10. browsers

eg: chrome, firefox,...

11. PC & Mobile games

12. Any kind of standalone applications